

CHEN, DANCHUN

B.S. Graduate in Cybersecurity

Fudan University, Shanghai, China

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EDUCATION

Fudan University

B.S. in Cybersecurity

College of Computer Science and Artificial Intelligence

Shanghai, China

Sep. 2020 - July. 2025

PUBLICATIONS

- Towards a Mechanistic Understanding of Propositional Logical Reasoning in Large Language Models
Danchun Chen, Qiyao Yan, Liangming Pan
arXiv preprint 2026. [Paper (earlier version)] [Code]
- Empowering Self-Verification of Small Language Models through Step-wise Reasoning and Verification
Danchun Chen, Yixin Cao, Liangming Pan
XLLM Workshop @ ACL 2025. 2nd Place in LLMSR. [Paper] [Code]

RESEARCH EXPERIENCE

Singapore Management University

Research Engineer, Full-time

Singapore

Apr. 2026 - Present

- Honored to be supervised by [Prof. Qianru Sun](#).

MOE Key Lab of Computational Linguistics, Peking University

Remote Intern (Mechanistic Interpretability)

Remote

Sep. 2025 - Mar. 2026

- Honored to be supervised by [Prof. Liangming Pan](#).
- Mechanistic Interpretability in Propositional Logic Reasoning (**EMNLP 2026, Under Review**)
 - Conducted mechanistic analysis on four models (Qwen3-8B/14B, Llama-3.1-8B-Instruct, Mistral-7B-Instruct) using *PropLogic-MI*, a controlled benchmark spanning 11 propositional logic rule categories across one- and two-hop reasoning.
 - Uncovered four interlocking mechanisms: Staged Computation, Information Transmission, Fact Retrospection, and Specialized Attention Heads, generalizable across models, rule categories, and reasoning hops.
 - Performed prompt-order control analysis confirming the staged organization is model-internal rather than input-layout-induced.

The Fudan University Natural Language Processing Group

Research Intern (LLM Reasoning)

Shanghai, China

Mar. 2025 - Jul. 2025

- Honored to be supervised by [Prof. Yixin Cao](#) and [Prof. Liangming Pan](#) (University of Arizona).
- Step-Wise Reasoning and Verification (SWRV) Framework (**XLLM @ ACL 2025, 2nd Place**)
 - Proposed SWRV, a two-stage Parser-Verifier framework for step-wise self-verification in small LMs (e.g., Llama-3-8B-Instruct) using minimal data.
 - Integrated deterministic symbolic solver Z3 by translating problems and reasoning steps into symbolic formulas for formal inference.
- Undergraduate Thesis: LLMs on LSAT Logic Games
 - Conducted systematic comparison of Long CoT and neuro-symbolic hybrid reasoning paradigms for LLMs on LSAT analytical reasoning; evaluated accuracy-efficiency trade-offs and performed data contamination detection.

Institute of Big Data, Fudan University

Research Engineer

Shanghai, China

Oct. 2022 - Mar.2023, Sept.2024 - Jun. 2025

- Honored to be supervised by [Prof. Yang Zhou](#).

- The Effect of Climate Change on Public Health (Sep. 2024 - Jun. 2025)
 - Merged high-resolution meteorological data with hospital visit records from 125 hospitals; employed fixed-effects models, CNNs and FASTNN to analyze temperature's influence on public health.
- How Much Can Machines Learn Finance from Chinese Text Data? (Oct. 2022 - Jun. 2023)
 - Implemented *Textual Factor Model* on Chinese stock market data. [\[Paper\]](#)

PROFESSIONAL EXPERIENCE

SAP Commerce Cloud

Full-Stack Software Engineer, **Full-time**

Shanghai, China

Nov. 2024 - Mar. 2026

- Implemented kubernetes-based multi-tenant SaaS platform for e-commerce configuration management with dynamic domain generation and tenant isolation.
- Led development of full-stack LLM proxy platform (internal) offering unified model access across multiple providers.
- Built agent workflows automating software release processes and internal operations using Dify and Claude Code (Skill/Plugin/Harness).

Poizon International Technology Department

Backend Software Engineering, **Intern**

Shanghai, China

Sep. 2023 - Feb. 2024

- Facilitated integration of European buy-and-sell platforms onto Dewu App; implemented product synchronization, attribute mapping, and expired product handling.
- Developed Feishu Robot for order query and SQL scripts for automated data consistency checks.

HONORS AND AWARDS

- **Certificate of 2nd Place**, ACL 2025 Workshop XLLM Shared Task: LLMSR 2025
- **National Third Prize**, National College Student Information Security Competition (Top 15% out of 774 teams) 2023
- **National Bronze Award**, China International College Students Innovation Competition (Top 0.4% out of 42,100,000 teams) 2023
- **Third Prize**, Shanghai College Students Mathematics Competition 2024
- **Undergraduate Excellence Scholarship**, Fudan University Multiple Years

LEADERSHIP EXPERIENCE

Fudan Runner Association

Founder And President

Shanghai, China

Sep. 2021 - Feb. 2023

- Led strategic planning and development for the association, set up the framework.
- Coordinated collaborations with Nike and secured initial funding for pre-registration operations.
- Organized over 20 events, each attracting a minimum of 15 participants, including three major races with over 100 attendees.
- Launched the *Run Bubble* campus event during the 2022 Shanghai lockdown, engaging 180+ participants and fostering student fitness enthusiasm amidst the pandemic.

SKILLS AND HOBBIES

- **Languages:** Chinese (Native), English (Fluent), Japanese (Elementary)
- **Programming:** Java, Python, C++, C, TypeScript, SQL, LaTeX, Linux
- **Sports:** Marathon, Hiking, Cycling
- **Travel:** Solo-traveled to half of mainland China and 10+ countries